

Module 1 — User & Authentication Management

ID	Test Scenario	Input	Expected Result
TC-01	Login with valid credentials	Correct email/password	User is logged in successfully
TC-02	Login with invalid password	Correct email + wrong password	Error message displayed
TC-03	Login with empty fields	Blank email/password	Validation message displayed
TC-04	Role-based login	Admin credentials + Admin role	Admin dashboard is displayed
TC-05	Unauthorized page access	Student tries Admin page	Access denied / redirected
TC-06	Logout	Click Logout	Session is terminated and login page shown
TC-07	Forgot password	Registered email	Password reset process initiated
TC-08	Invalid email format	abc.com	Email validation error displayed

Module 2 — Academic Master Data Management

Student Management

ID	Test Scenario	Input	Expected Result
TC-09	Add student	Valid student details	Student is added successfully
TC-10	Add student with missing data	Name/roll number blank	Validation error displayed
TC-11	Edit student	Change semester	Updated details saved
TC-12	Delete student	Select existing student	Student deleted successfully

TC-13	Search student	Enter roll number	Matching student displayed
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TC-14	Duplicate roll number	Existing roll number	Duplicate record rejected
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Subject/Faculty/Batch

ID	Test Scenario	Input	Expected Result
TC-15	Add subject	Valid subject details	Subject added
TC-16	Add faculty	Valid faculty details	Faculty added
TC-17	Create batch	Valid batch details	Batch created
TC-18	Assign faculty to subject	Faculty + Subject	Assignment saved
TC-19	Delete subject	Existing subject	Subject removed if no dependency exists

Module 3 — CO/PO Definition & Mapping

ID	Test Scenario	Input	Expected Result
TC-20	Add Course Outcome	CO1 + description	CO successfully created
TC-21	Add Programme Outcome	PO1 + description	PO successfully created
TC-22	Map CO to PO	Mapping strength = 3	Mapping saved as High
TC-23	No mapping	Strength = 0	CO-PO relationship stored as None
TC-24	Edit CO-PO mapping	Change 1 → 3	Updated mapping displayed

TC-25	Delete CO	Existing CO	CO deleted successfully
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TC-26	Invalid mapping value	Strength = 5	Invalid value rejected
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Module 4 — Assessment & Marks Entry

ID	Test Scenario	Input	Expected Result
TC-27	Create assessment	Valid assessment details	Assessment created
TC-28	Enter valid marks	Marks = 75/100	Marks saved successfully
TC-29	Marks greater than maximum	Marks = 110/100	Validation error
TC-30	Negative marks	Marks = -5	Validation error
TC-31	Empty marks	Blank marks	Appropriate validation displayed
TC-32	Upload valid CSV	Correct CSV file	Marks imported successfully
TC-33	Upload invalid CSV	Incorrect format	File rejected with error
TC-34	Update marks	Change 60 → 70	Updated marks stored

Module 5 — Aggregation & Attainment Engine

ID	Test Scenario	Input	Expected Result
TC-35	Calculate class average	Marks: 60, 70, 80	Average = 70
TC-36	Calculate highest marks	Marks: 60, 85, 75	Highest = 85
TC-37	Calculate lowest marks	Marks: 60, 85, 75	Lowest = 60
TC-38	Calculate pass percentage	Valid student marks	Correct pass % displayed
TC-39	Generate grade distribution	Student marks	Correct grades counted
TC-40	Calculate CO attainment	CO-wise marks	Correct CO attainment generated
TC-41	Calculate PO attainment	CO-PO mapping + CO attainment	Correct PO attainment generated
TC-42	No marks available	Empty assessment data	Appropriate message displayed

Module 6 — Dashboard & Visualization

ID	Test Scenario	Input	Expected Result
TC-43	Open Admin Dashboard	Admin login	Dashboard loads successfully
TC-44	Display student count	Student records exist	Correct student count displayed
TC-45	Display CO chart	CO attainment data	Chart displays correct values
TC-46	Performance trend	Multiple assessment records	Line chart displays trend
TC-47	Grade distribution	Grade data	Pie chart displays correct distribution

TC-48	Student dashboard	Student login	Personal academic information displayed
TC-49	Responsive layout	Mobile/tablet screen	UI adapts correctly
TC-50	Refresh dashboard data	New marks added	Updated analytics displayed

Module 7 — Report Generation & Export

ID	Test Scenario	Input	Expected Result
TC-51	Generate marksheet	Select student	Individual marksheet generated
TC-52	Generate CO report	Select subject	CO attainment report generated
TC-53	Generate PO report	Select batch/subject	PO report generated
TC-54	Generate class report	Select class	Class performance report generated
TC-55	Export PDF	Click PDF	PDF downloaded/generated
TC-56	Export Excel	Click Excel	Excel file generated
TC-57	Report with no data	Empty dataset	Appropriate message displayed